Experiment No. 5

Title

Understanding Virtual Terminals, Display Manager, X clients, and Window Managers

Aim

The problem is to explore and understand virtual terminals, display managers, X clients, and window managers, and to practice switching between text-based and graphical interfaces in a Linux environment.

Theory

Virtual Terminals

Linux supports multiple virtual terminals (VTs) that allow users to access several independent terminal sessions simultaneously. These terminals run in text mode and are useful for system recovery and troubleshooting. Common key combinations:

- Ctrl + Alt + F1 to Ctrl + Alt + F6 – Text-based virtual terminals
- Ctrl + Alt + F7 (or F1 on some systems) – Graphical interface

Display Manager

A display manager provides the graphical login interface and manages user sessions. It starts the graphical server and handles authentication. Common display managers:

- GDM (GNOME Display Manager)
- LightDM
- SDDM X

Window System

The X Window System (X11) is responsible for handling graphical display, keyboard, and mouse input. It enables graphical applications (X clients) to run on a Linux system. X Clients X clients are graphical applications that communicate with the X server to display windows. Examples include:

- Terminal emulator
- Web browser
- Text editor

Window Manager

A window manager controls the placement, appearance, and behavior of windows on the screen. Examples

- GNOME Shell
- KDE Plasma
- Xfce
- Openbox

Commands Used

Commands	Description
tty	Display current terminal
who	Show logged-in users
w	Show user activity
startx	Start X Window System
systemctl status	Check display manager status
ps	View running processes
echo $DISPLAY	Display X session variable

Procedure / Steps

1) Log in to the Linux system and open the terminal.

2) Log in to the Linux system.

3) Open the terminal.

4) Check the current terminal:

- tty

```
┌──(setika㉿kali)-[~]
└─$ tty
/dev/pts/0

┌──(setika㉿kali)-[~]
└─$
```

5) to another virtual terminal using:

- Ctrl + Alt + F2

6) Log in and verify terminal access.

7) Switch back to graphical mode:

- Ctrl + Alt + F1 or Ctrl + Alt + F7

8) Check logged-in users

9) Check the display manager status:

- systemctl status gdm (Replace gdm with lightdm or sddm if applicable)

10) Verify X session variable:

- echo $DISPLAY

11) List running X-related processes:

- ps -e | grep XL

```
┌──(setika㉿kali)-[~]
└─$ echo $DISPLAY
:0.0

┌──(setika㉿kali)-[~]
└─$ ps -e | grep X
    809 tty7     00:04:48 Xorg
```

Outcomes

- ✓ Successful switching between virtual terminals.
- ✓ Identification of display manager and X session.
- ✓ Display of active user sessions.
- ✓ Confirmation of graphical environment variables.

Result

- Virtual terminals, display managers, X clients, and window managers were successfully explored. The student gained practical understanding of Linux system interfaces and graphical session management.